

Mathematical Methods in Physics

Tensors and Linear Algebra — Prof. V. Balakrishnan

Handwritten lecture notes, transcribed

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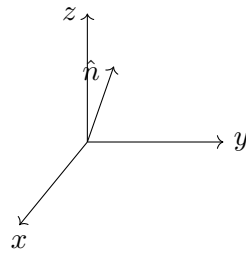
1 Chapter 1: Tensors

(Tensors)

In an n -dimensional space a tensor of rank ' r ' is a mathematical object that has ' r ' indices and n^r components and obey certain transformation rules.

[the word "components" is written in below the line, as an insertion]

components in a given basis :-



$$\vec{A} = x\hat{i} + y\hat{j} + z\hat{k} \quad \text{here } (\vec{A} = \vec{r})$$

components in basis $(\hat{i}, \hat{j}, \hat{k})$

vector can also define area ,

$$d\vec{A} = |dA| \hat{n}$$



(area = dA)

scalar

To specify temp of water



we don't need any basis vector.

0 Basis vector / component

vector :-

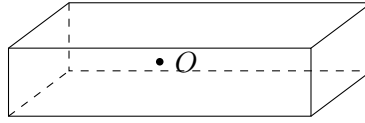
Per component we require one basis vector.

$$\vec{A} = 3\hat{i} + 4\hat{j}$$

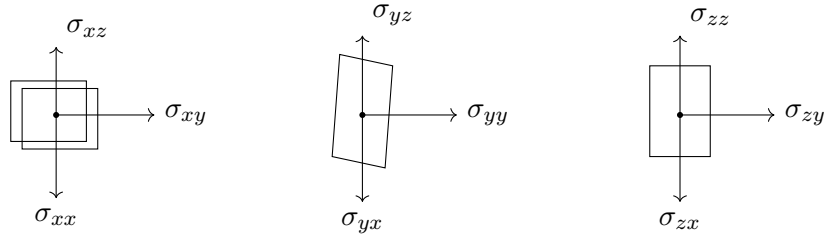
for each component we require only one basis vector.

Tensor of rank 2 :- (2 basis vector / component)

How will you define stress at point O .



we can cut surface in 3 ways at point ' O '



first ' x ' indicate dirⁿ of area & second indices indicate dirⁿ of stress/pressure.

$$\sigma = \begin{bmatrix} \sigma_{xx} & \sigma_{xy} & \sigma_{xz} \\ \sigma_{yx} & \sigma_{yy} & \sigma_{yz} \\ \sigma_{zx} & \sigma_{zy} & \sigma_{zz} \end{bmatrix}$$

[we need 2 basis vectors to write one component.]

(arrows under the columns of the matrix) All components with σ along x -dirn | (σ along ' y ') | (σ along z)

[we can't add $\sigma_{xx} + \sigma_{yy} + \sigma_{zz}$ just because all are in same dirⁿ but these components have different nature]

longitudinal stress = σ_{xx}

shear stress = σ_{yx}

To specify piezoelectricity & stiffness we require 3 basis vector per component & 4 basis vector per component respectively.

So, For a tensor of rank ' r ' in ' n ' dimensional space has n^r components and require ' r ' basis to define one component.

Summary

scalar	—	Tensor of rank ' 0 '	($3^0 = 1$ component)
vector	—	Tensor of rank ' 1 '	($3^1 = 3$ components)
stress	—	Tensor of rank ' 2 '	($3^2 = 9$ components)
			($4^2 = 16$ components in 4-D ...)
piezoelectricity	—	Tensor of rank ' 3 '	(In 3D has $3^3 = 27$ components)
stiffness	—	Tensor of rank ' 4 '	(In 3D has $3^4 = 81$ components)

[the word "Tensor" is written and struck out on the line below "vector"]

In form of tensor, any quantity is represented as

$$T_i \quad \text{or} \quad T^i$$

component | basis vector (also called indices)

$$\{T^i = T \text{ super } i ; \quad T_i = T \text{ sub } i\}$$

Representation of Tensor quantity.

IMPORTANT POINTS

→ superscript represents contravariant tensor represented as column vector

$$p^i = (p^i)_{i=1,2,3} = \begin{pmatrix} p^1 \\ p^2 \\ p^3 \end{pmatrix}$$

→ A tensor quantity with subscript Represents covariant tensor with representation of row vector

$$L_i = (L_i)_{i=1,2,3} = (L_1 \quad L_2 \quad L_3)$$

→ A tensor of second rank is a matrix

$$M_i^j = \begin{bmatrix} M_1^1 & M_2^1 & M_3^1 \\ M_1^2 & M_2^2 & M_3^2 \\ M_1^3 & M_2^3 & M_3^3 \end{bmatrix}$$

→ Any twice repeated index in single term is summed over.

$$\sum_{i=1}^3 a_i x_i = a_1 x_1 + a_2 x_2 + a_3 x_3$$

$$\sum a_i x_i = a_i x_i \quad (\text{einstein summation convention})$$

$$a_{ij} x_j = a_{i1} x_1 + a_{i2} x_2 + a_{i3} x_3$$

$$a^i{}_i = a^1{}_1 + a^2{}_2 + a^3{}_3 \quad ; \quad a^i x_i = a^1 x_1 + a^2 x_2 + \cdots + a^n x_n$$

→ Any index repeated twice in a single term is called dummy index.

$$a_{ij} x_j \quad (j = \text{dummy index})$$

dummy index can be replaced with other index of same range.

$$a_{ij} x_j = a_{ik} x_k = a_{i\ell} x_\ell \neq a_{ii} x_i$$

range should be same $(j = 1, 2, 3, \quad k = 1, 2, 3, \quad \ell = 1, 2, 3)$

→ Any index occurring once in single term is called free index.

$$a_{ij} x_j \quad (i - \text{free index})$$

$(i = 1, 2, 3)$

index ' i ' can have one out of 3 values.

$$a_{ij}x_j = a_{1j}x_j = a_{2j}x_j = a_{3j}x_j$$

here

$$a_{1j}x_j = a_{11}x_1 + a_{12}x_2 + a_{13}x_3$$

$$a_{2j}x_j = a_{21}x_1 + a_{22}x_2 + a_{23}x_3$$

$$a_{3j}x_j = a_{31}x_1 + a_{32}x_2 + a_{33}x_3$$

$$\boxed{a_{ij}x_j}$$

j = dummy index i = free index

- Repeats twice in single term. – occurs once in single term
- Represents sum over range – do not represent summation
- can be replaced by other index – can't be replaced by other index.
- Index can not occur more than 2 times.

$$\begin{cases} X_{ij}b_j & \checkmark \\ X_{ik}b_k & \checkmark \\ X_{ii}b_i & \times \end{cases}$$

while counting indices, count superscript & subscript together.

$$X^j_j \quad j = \text{dummy index}$$

$$X^j_j = * \text{ matrix (?)}$$

$$X_i^{jj} \quad \begin{array}{l} i = \text{free index} \\ j = \text{dummy index} \end{array}$$

→ In an equation involving einstein notation free indices on both sides must match

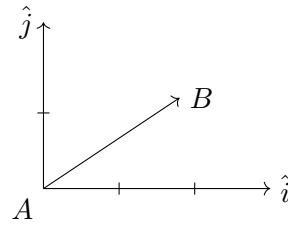
- (1) $x_i = a_{ij}b_j \quad \checkmark$
- (2) $x_i = a_{jj} \quad (\times) \quad (\text{invalid eq}^n)$
- (3) $a_i = A_{ki}B_{kj}x_j \quad (\checkmark)$
- (4) $x_j = A_{ik}v_k \quad (\times) \quad (\text{invalid eq}^n)$

– Contravariant Tensor & covariant tensor

Tensor $\vec{A}\vec{B}$ remains unchanged under general rotation about origin.

In general, in coordinate transformations component of tensor transforms in two manner contravariant and covariant.

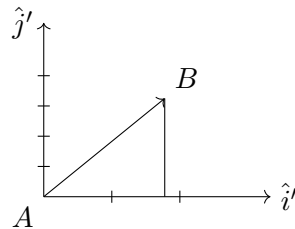
Contravariant tensor :-



$$\vec{AB} = 3\hat{i} + 2\hat{j}$$

$$\hat{i}' = 2\hat{i}$$

$$\hat{j}' = \hat{j}$$



$$\vec{AB} = 1.5\hat{i}' + 3\hat{j}'$$

components of tensor $\vec{AB}$ are changing inverse of change of basis vectors.

so, we call these tensor as contravariant tensor.

If Basis is doubled $\rightarrow$ component halved.

We represent such quantities as

$$x^i, v^i, a^i \dots$$

displacement, velocity, acceleration are contravariant vectors.

covariant tensor :-

A tensor whose components change in same manner as basis change under given transformation.

ex

$$\nabla F = \frac{\partial F}{\partial x}\hat{i} + \frac{\partial F}{\partial y}\hat{j} + \frac{\partial F}{\partial z}\hat{k}$$

$$\hat{i}' = 2\hat{i}$$

$$\hat{j}' = 2\hat{j}$$

$$\hat{k}' = 2\hat{k}$$

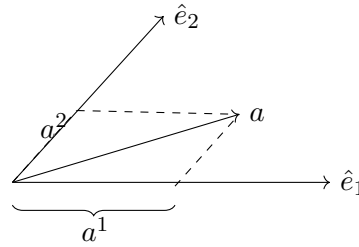
$\partial x =$ contravariant

$$\partial x' = \frac{\partial x}{2}, \quad \partial y' = \frac{\partial y}{2}, \quad \partial z' = \frac{\partial z}{2}$$

$$\nabla F = \frac{\partial F}{\partial x'} \hat{i}' + \frac{\partial F}{\partial y'} \hat{j}' + \frac{\partial F}{\partial z'} \hat{k}'$$

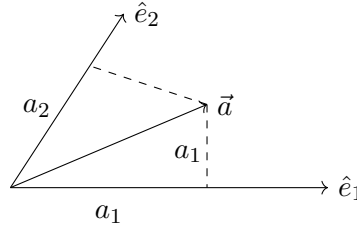
The quantities $\frac{\partial F}{\partial x'}$, $\frac{\partial F}{\partial y'}$, $\frac{\partial F}{\partial z'}$ also doubles just like basis vectors, such vectors are called as covariant tensors.

Tensor Notation :-



[If ' a' ' is contravariant vector, its components are calculated from parallel projection.]

$$\vec{a} = a^1 \hat{e}_1 + a^2 \hat{e}_2$$



[If $\vec{a}$ is covariant tensor, its components are calculated from perpendicular projection.]

$$\vec{a} = a_1 \hat{e}_1 + a_2 \hat{e}_2$$

Transformation law :-

(1) For covariant tensor :-

suppose $F(x)$ is scalar field defined in a coordinate system (x^i)

$$F(x) = F(x^1, x^2, x^3 \dots x^N)$$

$$\nabla F = \left(\frac{\partial F}{\partial x^1}, \frac{\partial F}{\partial x^2}, \frac{\partial F}{\partial x^3} \dots \frac{\partial F}{\partial x^n} \right)$$

where $u_i = \frac{\partial F}{\partial x^i}$ ($i = 1, 2, 3 \dots n$) ($1 \leq i \leq n$)

$$u_i = \frac{\partial F}{\partial x^i}$$

represents i^{th} component of ∇F .

used as subsmpt as ∇F is covariant tensor.

$$\nabla F_i = \left(\frac{\partial F}{\partial x^i} \right) = u_i$$

If the coordinate system transformed to new coordinate system $(\bar{x}^i)$ gradient is defined as

$$\nabla F = \left(\frac{\partial F}{\partial \bar{x}^1}, \frac{\partial F}{\partial \bar{x}^2}, \frac{\partial F}{\partial \bar{x}^3} \cdots \frac{\partial F}{\partial \bar{x}^n} \right)$$

where,

$$\bar{u}_i = \frac{\partial F}{\partial \bar{x}^i} \quad (1 \leq i \leq N)$$

represents i^{th} component of ∇F .

$$\begin{aligned} \bar{u}_i &= \frac{\partial F}{\partial \bar{x}^i} = \frac{\partial F}{\partial \bar{x}^i} \left(\frac{\partial x^r}{\partial \bar{x}^i} \right) \\ &= \left(\frac{\partial F}{\partial x^r} \right) \frac{\partial x^r}{\partial \bar{x}^i} = \frac{\partial F}{\partial x^1} \frac{\partial x^1}{\partial \bar{x}^i} + \frac{\partial F}{\partial x^2} \frac{\partial x^2}{\partial \bar{x}^i} + \cdots \end{aligned}$$

$$\boxed{\bar{u}_i = u_r \frac{\partial x^r}{\partial \bar{x}^i}}$$

where $(u_r = \frac{\partial F}{\partial x^r})$

$$\bar{u}_i = u_r \frac{\partial x^r}{\partial \bar{x}^i} \quad \begin{array}{l} \bar{u}_i = \text{component of } \nabla F \text{ in } \bar{x}^i \\ u_r = \text{_____ in } x^i \end{array}$$

$\frac{\partial x^r}{\partial \bar{x}^i}$ represents transformation from $(x^i \text{ to } \bar{x}^i)$ coordinate system

Defⁿ :-

covariant 1st rank tensor :-

The vector field ' u ' is said to be covariant first rank tensor, if its component u_i in (x^i) coordinate system and $\bar{u}_i$ in $(\bar{x}^i)$ coordinate system are related by following Transformation.

$$\bar{u}_i = u_r \frac{\partial x^r}{\partial \bar{x}^i} \quad (1 \leq i \leq n)$$

(2) Transformation for contravariant tensor :-

suppose $v(x) = v(x^1, x^2 \cdots x^n)$ is velocity defined in coordinate system (x^i) $[(n\text{-d space})]$

$$\vec{v} = \left(\frac{\partial x^1}{\partial t}, \frac{\partial x^2}{\partial t}, \frac{\partial x^3}{\partial t}, \cdots \frac{\partial x^n}{\partial t} \right)$$

where

$$v^i = \frac{\partial x^i}{\partial t} \quad (1 \leq i \leq n)$$

If coordinate system is transformed to new coordinate system, $(\bar{x}^i)$

$$\begin{aligned}\vec{v} &= \left(\frac{\partial \bar{x}^1}{\partial t}, \frac{\partial \bar{x}^2}{\partial t}, \frac{\partial \bar{x}^3}{\partial t}, \dots \frac{\partial \bar{x}^n}{\partial t} \right) \\ \bar{v}^i &= \frac{\partial \bar{x}^i}{\partial t} \\ &= \frac{\partial \bar{x}^i}{\partial t} \frac{\partial x^r}{\partial x^r} \\ &= \frac{\partial \bar{x}^i}{\partial x^r} \left(\frac{\partial x^r}{\partial t} \right) = \frac{\partial \bar{x}^i}{\partial x^1} \frac{\partial x^1}{\partial t} + \frac{\partial \bar{x}^i}{\partial x^2} \frac{\partial x^2}{\partial t} + \dots \\ \boxed{\bar{v}^i = v^r \frac{\partial \bar{x}^i}{\partial x^r}} \quad v^r &= \frac{\partial x^r}{\partial t}\end{aligned}$$

(ex)

$$\bar{v}^1 = v^r \frac{\partial \bar{x}^1}{\partial x^r} = v^1 \frac{\partial \bar{x}^1}{\partial x^1} + v^2 \frac{\partial \bar{x}^1}{\partial x^2} + v^3 \frac{\partial \bar{x}^1}{\partial x^3} + \dots$$

Summary

$$\bar{v}^i = v^r \frac{\partial \bar{x}^i}{\partial x^r} \quad \bar{v}^i = \text{component of } \vec{v} \text{ in } \bar{x}^i \\ v^r = \text{_____ in } x^i$$

Defⁿ :- The vector field ' v ' is said to be contravariant 1st rank tensor if its components v^i in (x^i) coordinate system and $\bar{v}^i$ in $\bar{x}^i$ coordinate system are related by following set of transformation.

$$\bar{v}^i = v^r \frac{\partial \bar{x}^i}{\partial x^r} \quad (1 \leq i \leq n)$$

Summary :-

$$\begin{array}{cc} \underline{\text{covariant}} & \underline{\text{contravariant}} \\ \text{(in same manner)} & \text{(in opposite manner)} \end{array}$$

– Basis & components transform in same manner – Basis and components transform in opposite manner

– Indices are in subscript of component (T_i) – indices are superscript of tensor quantity (T^i)

$$\vec{a} = a_1 \hat{e}_1 + a_2 \hat{e}_2 \quad \vec{a} = a^1 \hat{e}_1 + a^2 \hat{e}_2$$

$$\text{ex } \nabla F)_i = u_i = \frac{\partial F}{\partial x^i} \quad \text{ex displacement, velocity, acceleration, } \dots$$

– Transformation rule – Transformation rule

$$\left\{ \bar{u}_i = u_r \frac{\partial x^r}{\partial \bar{x}^i} \right\} \quad \left\{ \bar{v}^i = v^r \frac{\partial \bar{x}^i}{\partial x^r} \right\}$$

(Q.1) expand $a_{ij}(b_i + c_j + d_k)$

(×)

$$\begin{cases} = a_{ij}b_i + a_{ij}c_j + a_{ij}d_k \\ = (a_{1j}b_1 + a_{2j}b_2 + a_{3j}b_3) + (a_{i1}c_1 + a_{i2}c_2 + a_{i3}c_3) + a_{ij}d_k \end{cases}$$

Brackets in Einstein notation

$$a_{ij}(b_i + c_j + d_k)$$

(step-1) Combine terms outside parantheses with each term inside seperately.

$$\begin{aligned} & (a_{ij}b_i) + (a_{ij}c_j) + (a_{ij}d_k) \\ & \begin{matrix} i = 2 & i = 1 & i = 1 \\ j = 1 & j = 2 & j = 1, \quad k > 1 \end{matrix} \\ & \left. \begin{matrix} i_{max} = 2 \\ j_{max} = 2 \end{matrix} \right\} i, j = \text{dummy} \\ & k_{max} = 1 \quad \longrightarrow \quad k = \text{free indices} \end{aligned}$$

Non- identities :-

(1)

$$a_{ij}(x_i + y_j) \neq a_{ij}x_i + a_{ij}y_j$$

$$LHS : a_{ij}(x_i + y_j) \quad \left. \begin{matrix} i_{max} = 2 \\ j_{max} = 2 \end{matrix} \right\} i, j = \text{dummys}$$

$$RHS \quad a_{ij}x_i + a_{ij}y_j$$

[under the first term: $i = \text{dummy}$, $j = \text{free}$; under the second term: $j = \text{dummy}$, $i = \text{free}$]

RHS is inconsistent with LHS

(2)

$$a_{ij}x_iy_j \neq a_{ij}x_jy_i \quad (i, j = \text{dummy both sides})$$

$$\text{but coeff of } x_1y_2 = \begin{cases} a_{12} & \text{LHS} \\ a_{21} & \text{RHS} \end{cases}$$

(3)

$$(a_{ij} + a_{ji})x_iy_i \neq 2a_{ij}x_iy_j$$

($i, j = \text{dummy both sides but}$) $(a_{ij} + a_{ji})$.

Identities :-

(1)

$$a_{ij}(x_j + y_j) = a_{ij}x_j + a_{ij}y_j$$

(2)

$$a_{ij}x_iy_j = a_{ij}y_jx_i$$

(3)

$$a_{ij}x_ix_j = a_{ji}x_ix_j \equiv a_{ji}x_jx_i$$

(4)

$$(a_{ij} + a_{ji})x_ix_j \equiv 2a_{ij}x_ix_j$$

(5)

$$(a_{ij} - a_{ji})x_ix_j \equiv 0$$

kronecker delta :-

$$\delta_{ij} \equiv \delta_j^i \equiv \delta^{ij} \equiv \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

$$\delta_{ij} = \delta_{ji} \quad (\forall i, j)$$

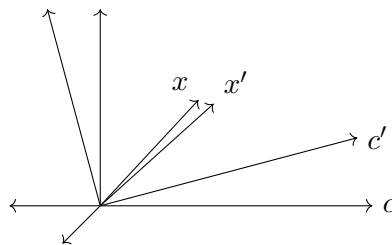
example Simplify $\delta_{ij}x_i x_j$

$$\delta_{ij}x_i x_j = \delta_{ii}x_i x_i = \delta_{jj}x_j x_j$$

$$\delta_{ij}x_i x_j = x_i x_i = x_j x_j \quad \checkmark$$

Linear Algebra of Tensors :-

linear transformations



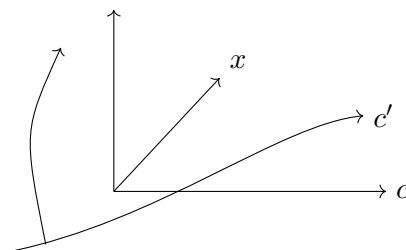
[linear – scaling, rotation]

Suppose we have vector ' x ' in coordinate system c , If we do linear transformation of coordinate system to c' , then vector x will look like x' in new c' coordinate system.

$$x' = L x$$

→ matrix (2×2) Transformation matrix

General Transformation :



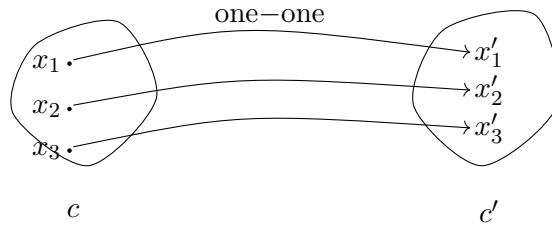
New vector will look like x' in new coordinate system c' ,

$$x' = f(x) = T(\vec{x})$$

(or)

$$x'_i = T_p(x_1, x_2, x_3 \cdots x_n)$$

A bijective transformation is one to one, every ' x ' in original coordinate system corresponds to unique x' in the image coordinate system & every x' in image coordinate system corresponds to a unique x in the original coordinate system.



The image of x' of bijective transformation ' T ' represents a set of admissible coordinates for x

$$\left(\begin{array}{cccc} \text{collection of } x_i & \text{represents } c \\ " & x'_i & " & c' \end{array} \right)$$

If T is linear x_i coordinate system is called affine coordinate system. If T is non-linear then $\{x_i\}$ makes up curvilinear coordinate systems ex, polar, cylindrical, spherical coordinate system.

Chain rule of partial derivative :-

Suppose

$$w = f(u_1, u_2, u_3 \cdots u_n)$$

where $u_i = u_i(x_1, x_2 \cdots x_m)$

Then we know that

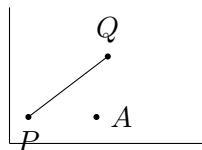
$$\frac{\partial w}{\partial x_j} = \frac{\partial f}{\partial u_1} \frac{\partial u_1}{\partial x_j} + \frac{\partial f}{\partial u_2} \frac{\partial u_2}{\partial x_j} + \frac{\partial f}{\partial u_3} \frac{\partial u_3}{\partial x_j} + \cdots$$

or

$$\left(\frac{\partial w}{\partial x_j} = \frac{\partial f}{\partial u_i} \frac{\partial u_i}{\partial x_j} \right) \leftarrow \text{einstiens summation convention}$$

Coordinate transformation for tensors :-

New notation :- A vector in $\mathbb{R}^n$



$$\overline{PQ} = \begin{pmatrix} x^1 \\ x^2 \\ x^3 \\ \vdots \\ x^N \end{pmatrix}$$

[use parantheses for power]

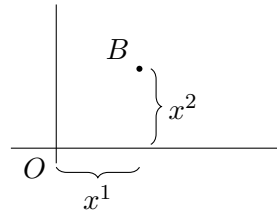
$$[(a_i)^2 = a_i a_i]$$

$$[(a^i)^2 = a^i \times a^i]$$

coordinate of 'A' in $\mathbb{R}^N =$

$$\begin{pmatrix} a^1 \\ a^2 \\ a^3 \\ \vdots \\ a^N \end{pmatrix}$$

Ractangular coordinates :-



$$B = \begin{pmatrix} x^1 \\ x^2 \end{pmatrix}$$

A coordinate system is said to be ractangular if distance b/w two points $C(x^1 x^2 \dots x^N)$ & $D(y^1 y^2 \dots y^N)$ is given by

$$\sqrt{(x^1 - y^1)^2 + (x^2 - y^2)^2 + (x^3 - y^3)^2 + \dots}$$

or

$$\sqrt{\delta_{ij} \Delta x^i \Delta x^j}$$

or

$$\sqrt{\Delta x^i \Delta x^i} \quad \Delta x^i = (x^i - y^i)$$

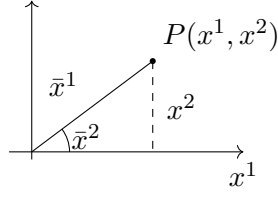
curvilinear coordinates :-

suppose:

- 1) P is a point in a coordinate system (x^i) in $\mathbb{R}^n$ given by $P : (x^1, x^2 \dots, x^n)$
- 2) $(\bar{x}^i)$ is another coordinate system in $\mathbb{R}^n$ such that the coordinates of ' p ' in this system are $(\bar{x}^1 \bar{x}^2 \dots \bar{x}^n)$
- 3) suppose that $\bar{x}^i = \bar{x}^i(x^1 x^2 x^3 \dots x^N)$ [F denotes N such transformations]. If functions $\bar{x}^i(x^1 x^2 \dots x^n)$ are all real valued, have continuous 2nd order partial derivatives every where and are all invertible then F is called coordinate transformation.

If (x^i) being inputs of F are ractangular coordinates then $(\bar{x}^i)$ is

- a) Affine if F is linear
 - b) curvilinear if F is not-linear
- examples of curvilinear coordinates
- (1) polar coordinates



$$\left. \begin{aligned} \bar{x}^1 &= \sqrt{(x^1)^2 + (x^2)^2} \\ \bar{x}^2 &= \arctan\left(\frac{x^2}{x^1}\right) \end{aligned} \right\} T_p$$

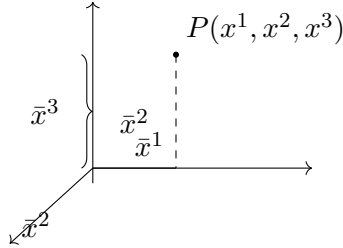
T_p is coordinate transformation

- As inputs of & outputs of T are real numbers
- T has or functions of T has continuous 2nd partial derivative everywhere (except origin)
- T_p is invertible

$$\left. \begin{aligned} x^1 &= \bar{x}^1 \cos(\bar{x}^2) \\ x^2 &= \bar{x}^1 \sin(\bar{x}^2) \end{aligned} \right\} T_p^{-1}$$

eqⁿ of T_p are non-linear so T_p represents curvilinear coordinate transformation

(2) cylindrical coordinates :

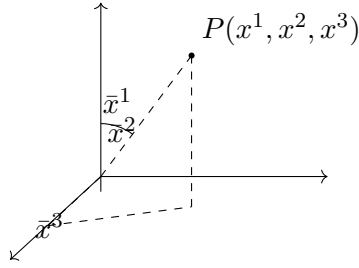


$$\left. \begin{aligned} \bar{x}^1 &= \sqrt{(x^1)^2 + (x^2)^2} \\ \bar{x}^2 &= \arctan\left(\frac{x^2}{x^1}\right) \\ \bar{x}^3 &= x^3 \end{aligned} \right\} T_p$$

$$T_p^{-1} \Rightarrow \begin{cases} x^1 = \bar{x}^1 \cos(\bar{x}^2) \\ x^2 = \bar{x}^1 \sin(\bar{x}^2) \\ x^3 = \bar{x}^3 \end{cases}$$

[Non-linear so represents Curvilinear Coordinate System.]

(3) spherical coordinate system :-



$$\left. \begin{aligned} \bar{x}^1 &= \sqrt{(x^1)^2 + (x^2)^2 + (x^3)^2} \\ \bar{x}^2 &= \cos^{-1} \left(\frac{x^3}{\sqrt{(x^1)^2 + (x^2)^2 + (x^3)^2}} \right) \\ \bar{x}^3 &= \tan^{-1} \left(\frac{x^2}{x^1} \right) \end{aligned} \right\} T_p$$

$$T_p^{-1} \Rightarrow \begin{cases} x^1 = \bar{x}^1 \sin(\bar{x}^2) \cos(\bar{x}^3) \\ x^2 = \bar{x}^1 \sin(\bar{x}^2) \sin(\bar{x}^3) \\ x^3 = \bar{x}^1 \cos(\bar{x}^2) \end{cases}$$

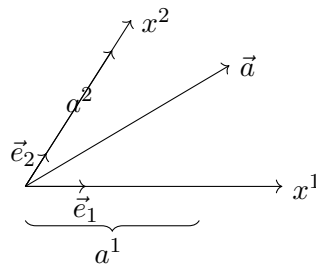
T_p & T_p^{-1} have non-linear functions so T_p & T_p^{-1} represents curvilinear coordinate transformation.

contravariant and covariant tensors :-

A Tensor whose components change in contravariant/covariant manner is said to be contravariant/Covariant tensor.

A) contravariant change :-

suppose we have 2-D non-orthogonal coordinate system.



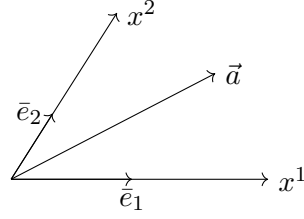
$$\vec{a} = a^1 \vec{e}_1 + a^2 \vec{e}_2$$

a^1 = no. of e_1 req.

a^2 = no. of e_2 required.

what happens to $\vec{a}$ if $e_1' = \alpha e_1$ & $e_2' = \beta e_2$

$\vec{a}$ does not change but its components will change



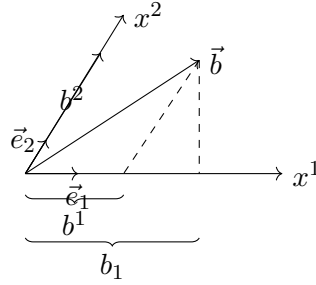
$$\vec{a} = \bar{a}^1 \bar{e}_1 + \bar{a}^2 \bar{e}_2$$

$$\bar{a}^1 = \frac{a^1}{\alpha} \quad \bar{a}^2 = \frac{a^2}{\beta}$$

So components $\bar{a}^1$ & $\bar{a}^2$ are changing in contrary to $\bar{e}_1$ & $\bar{e}_2$.

For representation purpose we use superscripts to distinguish from covariant quantities.

Covariant change :-



we can write $\vec{b}$ as $\vec{b} = b^1 e_1 + b^2 e_2$ in contravariant components.

We can also write $\vec{b} = b_1 e_1 + b_2 e_2$

where b_1 & b_2 are covariant components.

$$\vec{b} = b_1 \vec{e}_1 + b_2 \vec{e}_2$$

where

$$b_1 = \vec{b} \cdot e_1 = (b^1 e_1 + b^2 e_2) \cdot e_1 = b^1 (e_1 \cdot e_1) + b^2 (e_2 \cdot e_1)$$

$$b_2 = \vec{b} \cdot e_2 = (b^1 e_1 + b^2 e_2) \cdot e_2 = b^1 (e_1 \cdot e_2) + b^2 (e_2 \cdot e_2)$$

If we change our basis $e'_1 = \alpha e_1$ & $e'_2 = \beta e_2$ then

$$\vec{b} = b'_1 e'_1 + b'_2 e'_2$$

where

$$b'_1 = \vec{b} \cdot e'_1 = \vec{b} \cdot (\alpha e_1) = \alpha b_1$$

$$b'_2 = \vec{b} \cdot e'_2 = \vec{b} \cdot (\beta e_2) = \beta b_2$$

so we can see that components b'_1 & b'_2 also change in same manner as e'_1 & e'_2 are changing so such components are called as covariant components.

Caveats :-

- 1) vectors with covariant components are covectors (different from regular vectors)
 - 2) co-vectors are better thought as functions
 - 3) co-vectors are usually specified with dual vectors as basis instead of conventional basis vectors.
- *Note:* we have learned that contravariant vector 'v' transform under change of coordinate system ($x^i \rightarrow \bar{x}^i$) as

$$\bar{v}^i = v^r \frac{\partial \bar{x}^i}{\partial x^r} \quad 1 \leq i \leq N$$

ex

- Tangent vector of parametric curve ($x^1, x^2, x^3 \dots x^N$) (where $x^i = x^i(t)$)

$$\theta^i = \frac{dx^i}{dt}$$

is also contravariant vector / contravariant tensor of rank 1

under coordinate transformation $x^i \rightarrow \bar{x}^i$

$$\bar{\theta}^i = \frac{d\bar{x}^i}{dt} \quad \bar{x}^i = \bar{x}^i(x^1, x^2 \dots x^N)$$

$$\bar{\theta}^i = \frac{\partial \bar{x}^i}{\partial x^1} \frac{\partial x^1}{\partial t} + \frac{\partial \bar{x}^i}{\partial x^2} \frac{\partial x^2}{\partial t} + \dots$$

$$\bar{\theta}^i = \frac{\partial \bar{x}^i}{\partial x^r} \left(\frac{\partial x^r}{\partial t} \right)$$

$$\boxed{\bar{\theta}^i = \theta^r \frac{\partial \bar{x}^i}{\partial x^r}}$$

θ^i transform like contravariant tensor of rank 1 so it's an contravariant vector.

we have seen that covariant vector (Tensors of rank 1) transform $v_i \rightarrow \bar{v}_i$ under coordinate transformation $x^i \rightarrow \bar{x}^i$.

$$\bar{v}_i = v_r \frac{\partial x^r}{\partial \bar{x}^i}$$

To know how tensors of higher rank transform, In order to define higher rank tensors, we start with assumption.

[the word "components" is inserted above "tensors", and "tensor" is inserted above "rank"]

Contravariant and covariant tensors (formal definition) :-

Assumptions :- suppose that (w/V) is a matrix field (i.e. (w/V) provides us matrix for every point in space) composed of $n \times n$ scalar field (functions) defined

over region 'u' in $\mathbb{R}^n$.

Assume that in $\{x^i\}$ coordinate system, the components of V are V^{ij} . Assume also that after a coordinate transformation $T : \bar{x}^i = \bar{x}^i(x^1, x^2 \dots x^n)$ that takes us to the $\bar{x}^i$ coordinate system, the components of V becomes $\bar{V}^{ij}$.

matrix is not a tensor. A tensor is an object that is invariant under change of coordinate systems with components that change according to a special set of mathematical formulae

contravariant tensor (rank 2)

A matrix field V is a contravariant tensor of rank 2 if its components V^{ij} in $\{x^i\}$ coordinate system transform to $\bar{V}^{ij}$ in $\{\bar{x}^i\}$ coordinate system, such that

$$\bar{V}^{ij} = V^{rs} \frac{\partial \bar{x}^i}{\partial x^r} \frac{\partial \bar{x}^j}{\partial x^s}$$

$[i, j - \text{free indices}]$

$[r, s - \text{dummy indices.}]$

$$(1 \leq i, j, r, s \leq n)$$

covariant tensor (rank 2)

A matrix field 'W' is covariant tensor of rank-2 if its components W_{ij} in $\{x^i\}$ coordinate system & $\bar{W}_{ij}$ in $\{\bar{x}^i\}$ coordinate system obey

$$\bar{W}_{ij} = W_{rs} \frac{\partial x^r}{\partial \bar{x}^i} \frac{\partial x^s}{\partial \bar{x}^j}$$

Mixed tensor (rank -2)

suppose that Z is a matrix field composed of $n \times n$ scalar fields defined over region u in $\mathbb{R}^n$. Assume that in $\{x^i\}$ coordinate system, the components of Z are Z_j^i , which becomes $\bar{Z}_j^i$ after coordinate transformation $T : \bar{x}^i = \bar{x}^i(x^1, x^2, \dots, x^N)$

then Z (is matrix field) is mixed tensor of rank-2 of its components Z_j^i in x^i coordinate system. & $\bar{Z}_j^i$ in $\bar{x}^i$ coordinate system obey

$$\bar{Z}_j^i = Z_s^r \frac{\partial \bar{x}^i}{\partial x^r} \frac{\partial x^s}{\partial \bar{x}^j}$$

Tensor Ranks

Suppose V is tensor with components

$$V_{j_1 j_2 \dots j_q}^{i_1 i_2 i_3 \dots i_p}$$

then total rank of V is $(m = p + q)$.

we call V as (p, q) tensor.

$[p - \text{contravariant rank} ; q - \text{covariant rank.}]$

ex, If A is a tensor with components

$$A^{ij} \quad \text{then } A \text{ is } (2, 0) \text{ tensor}$$

A_{ij} then A is $(0, 2)$ tensor

A_j^i ——— (1, 1) tensor.

General Tensor

Assumption :- suppose V is an array field (It returns 1D/2D/3D – mD array) For each point in $\mathbb{R}^n$ composed of n^m scalar fields (functions) defined over a region 'u' in $\mathbb{R}^n$. Assume that in the (x^i) coordinate system components of V are $V_{j_1 j_2 \dots j_q}^{i_1 i_2 \dots i_p}$ where $(p + q = m)$. Assume also that after a coordinate transformation $T : \bar{x}^i = \bar{x}^i(x^1 x^2 \dots x^N)$ that takes us to $(\bar{x}^i)$ coordinate system components of $\bar{V}$ becomes.

$$\bar{V}_{j_1 j_2 \dots j_q}^{i_1 i_2 \dots i_p}$$

In order to be array field $\bar{V}$ to be tensor of rank $m = p + q$ with contravariant rank p & covariant rank 'q', then its components $\bar{V}_{j_1 j_2 \dots j_q}^{i_1 i_2 \dots i_p}$ in $(\bar{x}^i)$ coordinate system & $V_{j_1 j_2 \dots j_q}^{i_1 i_2 \dots i_p}$ in (x^i) coordinate system should obey

$$\bar{V}_{j_1 j_2 \dots j_q}^{i_1 i_2 \dots i_p} = V_{s_1 s_2 \dots s_q}^{r_1 r_2 \dots r_p} \frac{\partial \bar{x}^{i_1}}{\partial x^{r_1}} \frac{\partial \bar{x}^{i_2}}{\partial x^{r_2}} \frac{\partial \bar{x}^{i_3}}{\partial x^{r_3}} \dots \frac{\partial x^{s_1}}{\partial \bar{x}^{j_1}} \frac{\partial x^{s_2}}{\partial \bar{x}^{j_2}} \frac{\partial x^{s_q}}{\partial \bar{x}^{j_q}}$$

$$(1 \leq i, j, r, s \leq N)$$

[i's & j's are free indices]

[r's & s are dummy indices.]

Tensor operations

suppose that 'S' & 'T' are tensors given by

$$S \equiv \left(S_{j_1 j_2 \dots j_q}^{i_1 i_2 \dots i_p} \right), \quad T \equiv \left(T_{l_1 l_2 \dots l_q}^{k_1 k_2 \dots k_p} \right)$$

S is (p, q) tensor, T is (r, s) tensor.

1) summation :-

$$\left. \begin{matrix} p = r \\ q = s \end{matrix} \right\} \text{ contravariant \& covariant rank should be same.}$$

$$S + T = \left(S_{j_1 j_2 \dots j_q}^{i_1 i_2 \dots i_p} + T_{l_1 l_2 \dots l_q}^{k_1 k_2 \dots k_p} \right)$$

Then $(S + T)$ is tensor of rank same as S & T .

Proof :-

Assume that we change coordinates $x^i \rightarrow \bar{x}^i$

Then

$$\bar{S}_{j_1 j_2 \dots j_q}^{i_1 i_2 \dots i_p} = S_{\beta_1 \beta_2 \dots \beta_q}^{\alpha_1 \alpha_2 \dots \alpha_p} \frac{\partial \bar{x}^{i_1}}{\partial x^{\alpha_1}} \frac{\partial \bar{x}^{i_2}}{\partial x^{\alpha_2}} \dots \frac{\partial \bar{x}^{i_p}}{\partial x^{\alpha_p}} \frac{\partial x^{\beta_1}}{\partial \bar{x}^{j_1}} \frac{\partial x^{\beta_2}}{\partial \bar{x}^{j_2}} \dots \frac{\partial x^{\beta_q}}{\partial \bar{x}^{j_q}} \quad (1)$$

Similarly

$$\bar{T}_{j_1 j_2 \dots j_q}^{i_1 i_2 \dots i_p} = T_{\beta_1 \beta_2 \dots \beta_q}^{\alpha_1 \alpha_2 \dots \alpha_p} \frac{\partial \bar{x}^{i_1}}{\partial x^{\alpha_1}} \frac{\partial \bar{x}^{i_2}}{\partial x^{\alpha_2}} \dots \frac{\partial \bar{x}^{i_p}}{\partial x^{\alpha_p}} \frac{\partial x^{\beta_1}}{\partial \bar{x}^{j_1}} \frac{\partial x^{\beta_2}}{\partial \bar{x}^{j_2}} \dots \frac{\partial x^{\beta_q}}{\partial \bar{x}^{j_q}} \quad (2)$$

(1) $\pm$ (2) $\Rightarrow$

$$\bar{S}_{j_1 j_2 \dots j_q}^{i_1 i_2 \dots i_p} \pm \bar{T}_{j_1 j_2 \dots j_q}^{i_1 i_2 \dots i_p} = \left(S_{\beta_1 \beta_2 \dots \beta_q}^{\alpha_1 \alpha_2 \dots \alpha_p} \pm T_{\beta_1 \beta_2 \dots \beta_q}^{\alpha_1 \alpha_2 \dots \alpha_p} \right) C$$

$$C = \frac{\partial \bar{x}^{i_1}}{\partial x^{\alpha_1}} \frac{\partial \bar{x}^{i_2}}{\partial x^{\alpha_2}} \dots \frac{\partial \bar{x}^{i_p}}{\partial x^{\alpha_p}} \cdot \frac{\partial x^{\beta_1}}{\partial \bar{x}^{j_1}} \frac{\partial x^{\beta_2}}{\partial \bar{x}^{j_2}} \dots \frac{\partial x^{\beta_q}}{\partial \bar{x}^{j_q}}$$

[$\downarrow$ Transformation law for $(S \pm T)$]

$S + T$ is also tensor of same rank as S & T (p, q)

$$\left(\begin{array}{l} S, T \text{ are } (p, q) \text{ tensor} \\ (S \pm T) \text{ is } (p, q) \text{ tensor} \end{array} \right)$$

Scalar multiplication

(2) Scalar multiplication :-

Suppose A is a (p, q) tensor. If ϕ is a scalar then ϕA is found by

$$\phi A_{j_1 j_2 \dots j_q}^{i_1 i_2 \dots i_p}$$

i.e. multiplying each component $A_{j_1 j_2 \dots j_q}^{i_1 i_2 \dots i_p}$ by ϕ .

Then ϕA is also (p, q) tensor.

Proof :- A transform like - (If it is a (p, q) tensor)

$$\bar{A}_{j_1 j_2 \dots j_q}^{i_1 i_2 \dots i_p} = A_{\beta_1 \beta_2 \dots \beta_q}^{\alpha_1 \alpha_2 \dots \alpha_p} \left(\frac{\partial \bar{x}^{i_1}}{\partial x^{\alpha_1}} \frac{\partial \bar{x}^{i_2}}{\partial x^{\alpha_2}} \dots \frac{\partial \bar{x}^{i_p}}{\partial x^{\alpha_p}} \right) \left(\frac{\partial x^{\beta_1}}{\partial \bar{x}^{j_1}} \frac{\partial x^{\beta_2}}{\partial \bar{x}^{j_2}} \dots \frac{\partial x^{\beta_q}}{\partial \bar{x}^{j_q}} \right)$$

we can multiply each side by ϕ (scalar)

$$\phi \bar{A}_{j_1 j_2 \dots j_q}^{i_1 i_2 \dots i_p} = \phi A_{\beta_1 \beta_2 \dots \beta_q}^{\alpha_1 \alpha_2 \dots \alpha_p} \left(\frac{\partial \bar{x}^{i_1}}{\partial x^{\alpha_1}} \frac{\partial \bar{x}^{i_2}}{\partial x^{\alpha_2}} \dots \frac{\partial \bar{x}^{i_p}}{\partial x^{\alpha_p}} \right) \left(\frac{\partial x^{\beta_1}}{\partial \bar{x}^{j_1}} \frac{\partial x^{\beta_2}}{\partial \bar{x}^{j_2}} \dots \frac{\partial x^{\beta_q}}{\partial \bar{x}^{j_q}} \right)$$

since ϕ is scalar ($\bar{\phi} = \phi$)

$$\bar{\phi} \bar{A}_{j_1 j_2 \dots j_q}^{i_1 i_2 \dots i_p} = \phi A_{\beta_1 \beta_2 \dots \beta_q}^{\alpha_1 \alpha_2 \dots \alpha_p} \left(\frac{\partial \bar{x}^{i_1}}{\partial x^{\alpha_1}} \frac{\partial \bar{x}^{i_2}}{\partial x^{\alpha_2}} \dots \frac{\partial \bar{x}^{i_p}}{\partial x^{\alpha_p}} \right) \left(\frac{\partial x^{\beta_1}}{\partial \bar{x}^{j_1}} \frac{\partial x^{\beta_2}}{\partial \bar{x}^{j_2}} \dots \frac{\partial x^{\beta_q}}{\partial \bar{x}^{j_q}} \right)$$

So this implies (ϕA) also obey transformation law of that of A i.e. (p, q) tensor. So (ϕA) is also a (p, q) tensor.

Linear combinations

(3) Linear combinations :-

If $(T_1 \ T_2 \ T_3 - T_u)$ are tensors of same contravariant & covariant rank i.e. (p, q) tensor then, $(\lambda_1 T_1 + \lambda_2 T_2 + \lambda_3 T_3 - \lambda_u T_u)$ is also tensor of same contravariant & covariant rank. i.e.

$$(\lambda_1 T_1 + \lambda_2 T_2 + \dots \lambda_u T_u) = \lambda_i T_i \text{ is also } (p, q) \text{ tensor.}$$

$$(\text{if } T_i \text{ is } (p, q) \text{ tensor}).$$

$$(1 \leq i \leq u).$$

Contraction

Tensor operations :-

suppose 'S' & 'T' are tensors given by:

$$S \text{ is } (p, q) \text{ tensor} : S = \left(S_{j_1 j_2 \dots j_q}^{i_1 i_2 \dots i_p} \right)$$

$$T \text{ is } (r, s) \text{ tensor} : T = \left(T_{l_1 l_2 \dots l_s}^{k_1 k_2 \dots k_r} \right)$$

i.e

$$\bar{S}_{j_1 j_2 \dots j_q}^{i_1 i_2 \dots i_p} = S_{\beta_1 \beta_2 \dots \beta_q}^{\alpha_1 \alpha_2 \dots \alpha_p} \left(\frac{\partial \bar{x}^{i_1}}{\partial x^{\alpha_1}} \frac{\partial \bar{x}^{i_2}}{\partial x^{\alpha_2}} \dots \frac{\partial \bar{x}^{i_p}}{\partial x^{\alpha_p}} \right) \left(\frac{\partial x^{\beta_1}}{\partial \bar{x}^{j_1}} \frac{\partial x^{\beta_2}}{\partial \bar{x}^{j_2}} \dots \frac{\partial x^{\beta_q}}{\partial \bar{x}^{j_q}} \right)$$

$$\bar{T}_{l_1 l_2 \dots l_s}^{k_1 k_2 \dots k_r} = T_{b_1 b_2 \dots b_s}^{a_1 a_2 \dots a_r} \left(\frac{\partial \bar{x}^{k_1}}{\partial x^{a_1}} \frac{\partial \bar{x}^{k_2}}{\partial x^{a_2}} \dots \frac{\partial \bar{x}^{k_r}}{\partial x^{a_r}} \right) \left(\frac{\partial x^{b_1}}{\partial \bar{x}^{l_1}} \frac{\partial x^{b_2}}{\partial \bar{x}^{l_2}} \dots \frac{\partial x^{b_s}}{\partial \bar{x}^{l_s}} \right)$$

Contraction :-

suppose S is (p, q) tensor such that

$$S = \left(S_{j_1 j_2 \dots j_q}^{i_1 i_2 \dots i_p} \right)$$

The contraction of S w.r.t a contravariant index i_f and covariant index j_g (where $(1 \leq f \leq p)$ & $(1 \leq g \leq q)$) is given by setting $(i_f = j_g = u)$ and evaluating

$$(S' \equiv S_{j_1 j_2 \dots u \dots j_q}^{i_1 i_2 \dots u \dots i_p})$$

[S' is $(p-1, q-1)$ tensor]

example :- Suppose we have $(2, 2)$ tensor $A = A_{j_1 j_2}^{i_1 i_2}$

contraction of A relative to i_2 & j_1 ($A' : A_{u j_2}^{i_1 u}$)

A' = contraction of A w.r.t i_2 & j_1

$$A' = A_{u, j_2}^{i, u}$$

[$\leftarrow$ (repeated index gets summed over)]

in $\mathbb{R}^n$,

$$A' = A_{1, j_2}^{i, 1} + A_{2, j_2}^{i, 2} + A_{3, j_2}^{i, 3} + \cdots + A_{n, j_2}^{i, n}$$

in 3-D ($\mathbb{R}^3$,

$$A' = A_{1, j_2}^{i, 1} + A_{2, j_2}^{i, 2} + A_{3, j_2}^{i, 3}$$

[(1, 1) tensor]

$\therefore$ in going from $A \longrightarrow A'$

contravariant & covariant rank reduces by 1.

So A' is (1, 1) tensor.

Proof :-

For S is a (p, q) tensor, we know its transformation law,

$$\bar{S}_{j_1 j_2 \dots j_q}^{i_1 i_2 \dots i_p} = S_{\beta_1 \beta_2 \dots \beta_q}^{\alpha_1 \alpha_2 \dots \alpha_p} \frac{\partial \bar{x}^{i_1}}{\partial x^{\alpha_1}} \frac{\partial \bar{x}^{i_2}}{\partial x^{\alpha_2}} \cdots \frac{\partial \bar{x}^{i_p}}{\partial x^{\alpha_p}} \frac{\partial \bar{x}^{j_1}}{\partial x^{\beta_1}} \frac{\partial \bar{x}^{j_2}}{\partial x^{\beta_2}} \cdots \frac{\partial \bar{x}^{j_q}}{\partial x^{\beta_q}} \quad (1)$$

$$(i_f = j_g = u)$$

$$\bar{S}_{j_1 j_2 \dots j_q}^{i_1 i_2 \dots i_p} = S_{\beta_1 \beta_2 \dots \beta_q}^{\alpha_1 \alpha_2 \dots \alpha_p} \left(\frac{\partial \bar{x}^{i_1}}{\partial x^{\alpha_1}} \frac{\partial \bar{x}^{i_2}}{\partial x^{\alpha_2}} \cdots \frac{\partial \bar{x}^{i_p}}{\partial x^{\alpha_p}} \right) \left(\frac{\partial \bar{x}^u}{\partial x^{\alpha_f}} \frac{\partial x^{\beta_g}}{\partial \bar{x}^u} \right) \left(\frac{\partial x^{\beta_1}}{\partial \bar{x}^{j_1}} \frac{\partial x^{\beta_2}}{\partial \bar{x}^{j_2}} \cdots \frac{\partial x^{\beta_q}}{\partial \bar{x}^{j_q}} \right) \quad (2)$$

we know that from reverse chain rule

$$\left(\frac{\partial \bar{x}^u}{\partial x^{\alpha_f}} \frac{\partial x^{\beta_g}}{\partial \bar{x}^u} \right) = \frac{\partial x^{\beta_g}}{\partial x^{\alpha_f}} = \delta_{\alpha_f}^{\beta_g}$$

$$\delta_{\alpha_f}^{\beta_g} = \begin{cases} 1 & \beta_g = \alpha_f \\ 0 & \beta_g \neq \alpha_f \end{cases}$$

So, eqn (2) becomes transformation law for $(p-1, q-1)$ tensor

so S' represents $(p-1, q-1)$ tensor.

Inner product

(2) Inner product :-

suppose S and T are two tensors given by

$$S = \left(S_{j_1 j_2 \dots j_q}^{i_1 i_2 \dots i_p} \right) \quad \text{and} \quad T = \left(T_{l_1 l_2 \dots l_s}^{k_1 k_2 \dots k_r} \right)$$

To find inner product $S \cdot T$, choose a contravariant index on S (i_f) and covariant index (l_g) on T .

let ($i_f = l_g = u$) then

$$S \circ T = \left(S_{j_1 j_2 \dots j_q}^{i_1 i_2 \dots u \dots i_p} T_{l_1 l_2 \dots u \dots l_s}^{k_1 k_2 \dots k_r} \right)$$

$S \cdot T$ is $(p + r - 1, q + s - 1)$ tensor

or total rank of $S \cdot T = m = p + r - 1 + q + s - 1$

$$= p + r + q + s - 2$$

Proof can be found similer previous method.

ex (1) Evaluate $u \cdot v$ where, $V = (V^i)$ and $u = (u_j)$

$$V \cdot u = V^s u_s \quad (i = j = s)$$

i.e., (in $\mathbb{R}^3$)

$$V \cdot u = V^1 u_1 + V^2 u_2 + V^3 u_3$$

V, u & tensors following

$$\bar{V}^i = V^r \frac{\partial \bar{x}^i}{\partial x^r}$$

$$\bar{u}_j = u_s \frac{\partial x^s}{\partial \bar{x}^j}$$

for inner product ($i = j = k$)

$$\bar{V}^k \bar{u}_k = \left(V^r \frac{\partial \bar{x}^k}{\partial x^r} \right) \left(u_s \frac{\partial x^s}{\partial \bar{x}^k} \right)$$

$$= V^r u_s \frac{\partial \bar{x}^k}{\partial x^r} \frac{\partial x^s}{\partial \bar{x}^k}$$

$$= V^r u_s \frac{\partial x^s}{\partial x^r}$$

$$= V^r u_s \delta_r^s$$

$$\left[\frac{\partial x^s}{\partial x^r} = \delta_r^s \right]$$

$$\left(\bar{V}^k \bar{u}_k = V^s u_s \right)$$

inner product of contravariant and covariant vector is invariant / scalar.

$$m = \text{rank of } S \cdot T \mid V \cdot u = p + q + r + s - 2$$

$$= 1 + 1 - 2 = \underline{0}$$

So $(V \cdot u = \text{scalar})$ *i.e.* tensor of rank $\underline{0}$.

Outer product

(3) Outer product :-

Suppose S & T are tensors given by

$$S = \left(S_{j_1 j_2 \dots j_q}^{i_1 i_2 \dots i_p} \right) \quad \& \quad T = \left(T_{l_1 l_2 \dots l_s}^{k_1 k_2 \dots k_r} \right)$$

'S' is (p, q) tensor & 'T' is (r, s) tensor.

Then the outer product / direct product / tensor product

$$S \otimes T \equiv \left(S_{j_1 j_2 \dots j_q}^{i_1 i_2 \dots i_p} \cdot T_{l_1 l_2 \dots l_s}^{k_1 k_2 \dots k_r} \right)$$

$S \otimes T$ is tensor whose components are product of each component of S & each component of T .

$S \otimes T$ is $(p + r, q + s)$ tensor.

$S \otimes T$ is tensor of rank $(m = p + r + q + s)$.

example (1) Find outer product of $a = \begin{pmatrix} a^1 \\ a^2 \\ a^3 \end{pmatrix}$ & $b = (b_1 \quad b_2 \quad b_3)$

$$a \otimes b = \begin{pmatrix} a^1 b_1 & a^1 b_2 & a^1 b_3 \\ a^2 b_1 & a^2 b_2 & a^2 b_3 \\ a^3 b_1 & a^3 b_2 & a^3 b_3 \end{pmatrix}$$

[$\rightarrow$ mixed tensor of rank 2.]

$\downarrow$

$$a^i \otimes b_j$$

(Chapter -1) The Einstein summation convention

[(Schaum's outlines)]

(1) (i)

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = \sum_{i=1}^N a_ix_i \equiv a_ix_i$$

(Repeated indices gets summed over)

(ii)

$$a_{ii}x_K = a_{11}x_K + a_{22}x_K + a_{33}x_K + \cdots + a_{nn}x_n$$

(iii)

$$a_{ij}x_j = a_{i1}x_1 + a_{i2}x_2 + a_{i3}x_3 + \cdots + a_{in}x_n$$

repeated indices = dummy indices

non-repeated indices = free indices.

Note

$$a_{ij}x_j \neq a_{kj}x_j \quad (\text{unless } i = k)$$

(iv)

$$y_i = a_{ir}x_r$$

$$y_i = a_{i1}x_1 + a_{i2}x_2 + a_{i3}x_3$$

[tensor of rank 1]

[If $(y_i \in \mathbb{R}^3)$]

$$y_1 = a_{11}x_1 + a_{12}x_2 + a_{13}x_3$$

$$y_2 = a_{21}x_1 + a_{22}x_2 + a_{23}x_3$$

$$y_3 = a_{31}x_1 + a_{32}x_2 + a_{33}x_3$$

$(x_j)(y_i)$ = covariant tensor of rank 1

a_{ir} = covariant tensor of rank 2

$$\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

$$\boxed{y_i = a_{ir}x_r}$$

Einstein summation convention

Any expression involving a twice repeated index (occurring twice as superscript, or twice as subscript, or once a superscript and once a subscript) shall automatically stand for its sum over the values $1, 2, 3, \dots, n$. of repeated index.

$$a_i^i = a_1^1 + a_2^2 + a_3^3 + \dots + a_n^n$$

for a_j^i to be $(1, 1)$ tensor in n-D. ($\mathbb{R}^n$)

$$a_{ii} = a_{11} + a_{22} + a_{33} + \dots + a_{nn}$$

$$a^{jj} = a^{11} + a^{22} + a^{33} + \dots + a^{nn}$$

$$a^i x^i = a^1 x^1 + a^2 x^2 + \dots + a^n x^n$$

If $a^i x^i$ is an tensor of rank 0. (scalar).

Substitutions

$$Q = b_{ij}y_i x_j \quad \text{where } y_i = a_{ij}x_j$$

But 'j' is duplicated so

we replace $j \rightarrow r$ (different index) to substitute y_i in Q .

$$Q = b_{ij}y_i x_j = b_{ij} (a_{ir}x_r) x_j$$

$$Q = a_{ir}b_{ij}x_r x_j$$

ex If $y_i = a_{ij}x_j$ then express quadratic form

$$Q = g_{ij}y_i y_j \quad \text{in terms of } x - \text{variables.}$$

$$Q = g_{ij}y_i y_j$$

where $y_i = a_{ir}x_r$ (j is replaced with r) as j is duplicated

$$y_j = a_{js}x_s$$

$$Q = g_{ij} (a_{ir}x_r) (a_{js}x_s) = (g_{ij}a_{ir}a_{js}) x_r x_s$$

(or)

$$Q = h_{rs}x_r x_s$$

where

$$h_{rs} = g_{ij}a_{ir}a_{js}$$

(i & j gets summed over).

(1.3) kronecker delta and algebraic manipulations

A much used symbol in tensor calculus has the effect of annihilating the off-diagonal terms in a double summation.

kronecker delta

$$\delta_{ij} = \delta^{ij} = \delta_j^i = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

clearly ($\delta_{ij} = \delta_{ji} \quad \forall i \text{ \& } j$).

ex (1)

$$\delta_{ij}x_i x_j = ? \quad \text{for } 1 \leq i, j \leq 3$$

$$\delta_{ij}x_i x_j = \delta_{i1}x_i x_1 + \delta_{i2}x_i x_2 + \delta_{i3}x_i x_3$$

$$= \left(\delta_{11}x_1x_1 + \underbrace{\delta_{21}x_2x_1}_0 + \underbrace{\delta_{31}x_3x_1}_0 \right) + \left(\delta_{22}x_2x_2 + \underbrace{\delta_{12}x_1x_2}_0 + \underbrace{\delta_{32}x_3x_2}_0 \right) + \left(\underbrace{\delta_{13}x_1x_3}_0 + \underbrace{\delta_{23}x_2x_3}_0 + \delta_{33}x_3x_3 \right)$$

$$\delta_{ij}x_i x_j = x_1^2 + x_2^2 + x_3^2 = x_i x_i = x_j x_j$$

$$\delta_j^r a_{ir} x_i = a_{ij} x_i$$

$$\delta_j^r = \begin{cases} 1 & r = j \\ 0 & r \neq j \end{cases}$$

ex 2) Suppose $T^i = g_r^i a_{rs} y_s$, $y_i = b_{ir} x_r$, $a_{ir} b_{rj} = \delta_{ij}$

Find T^i in terms of x_p .

$$T^i = g_r^i a_{rs} y_s$$

$[y_s = b_{sr} x_r \text{ but } (r \text{ is duplicated}) \text{ so } y_s = b_{sj} x_j]$

$$T^i = g_r^i a_{rs} b_{sj} x_j$$

$$= g_r^i \delta_{rj} x_j$$

$$(T^i = g_r^i x_r)$$

Non-identities

$$a_{ij} (x_i + y_j) \neq a_{ij} x_i + a_{ij} y_j$$

$$a_{ij} x_i y_j \neq a_{ij} y_i x_j$$

$$(a_{ij} + a_{ji}) x_i y_j \neq 2a_{ij} x_i x_j$$

Problems

Repeated indices :-

use summation convention to write following

(a)

$$a_{11} b_{11} + a_{21} b_{12} + a_{31} b_{13} + a_{41} b_{14} = a_{i1} b_{1i}$$

$a_{i1} b_{1i}$ is tensor of rank 1 in $\mathbb{R}^4$. ($n = 4$)

(b)

$$a_{11} b_{11} + a_{12} b_{12} + a_{13} b_{13} + a_{14} b_{14} + a_{15} b_{15} + a_{16} b_{16}$$

$$= a_{1i} b_{1i} \quad \text{in } \mathbb{R}^6 \quad (n = 6)$$

(c)

$$C_{11}^i + C_{22}^i + C_{33}^i + C_{44}^i + C_{55}^i + C_{66}^i + C_{77}^i + C_{88}^i = C_{jj}^i$$

Tensor in $\mathbb{R}^8$ ($n = 8$)

Exercise 1.2

(1.2) use summation convention to write following systems.

(a)

$$\left. \begin{aligned} C_{11}x_1 + C_{12}x_2 + C_{13}x_3 &= 2 \\ C_{21}x_1 + C_{22}x_2 + C_{23}x_3 &= -3 \\ C_{31}x_1 + C_{32}x_2 + C_{33}x_3 &= 5 \end{aligned} \right\} \quad \begin{aligned} C_{1j}x_j &= 2 \\ C_{2j}x_j &= -3 \\ C_{3j}x_j &= 5 \end{aligned}$$

$$\boxed{C_{ij}x_j = d_i \quad \text{where } (d_1 = 2, d_2 = -3, d_3 = 5)}$$

(b)

$$a_j^1x_1 + a_j^2x_2 + a_j^3x_3 + a_j^4x_4 = b_j \quad (j = 1, 2)$$

$$a_j^i x_i = b_j$$

[j = free index ; i = dummy index]

[$j = 1, 2$; $i = 1, 2, 3, 4$]

Exercise 1.3

(1.3) write out explicit summations

1) $C_i(x_i + y_i)$ & $C_jx_j + C_ky_k$ ($n = 4$)

$$C_i(x_i + y_i) = C_1(x_1 + y_1) + C_2(x_2 + y_2) + C_3(x_3 + y_3) + C_4(x_4 + y_4)$$

$$= (C_1x_1 + C_2x_2 + C_3x_3 + C_4x_4) + (C_1y_1 + C_2y_2 + C_3y_3 + C_4y_4)$$

$$= C_jx_j + C_ky_k$$

$$\boxed{C_i(x_i + y_i) = C_jx_j + C_ky_k}$$

Double sum

Exercise 1.4

(1.4) If $n = 3$, expand $Q = a^{ij}x_ix_j$

$$Q = a^{1j}x_1x_j + a^{2j}x_2x_j + a^{3j}x_3x_j$$

$$= (a^{11}x_1x_1 + a^{12}x_1x_2 + a^{13}x_1x_3) + (a^{21}x_2x_1 + a^{22}x_2x_2 + a^{23}x_2x_3) + (a^{31}x_3x_1 + a^{32}x_3x_2 + a^{33}x_3x_3)$$

Exercise 1.5

(1.5) use summation convention to write the following & state $n = ?$

(a)

$$(a_{11}b_{11} + a_{21}b_{12} + a_{31}b_{13}) + (a_{12}b_{21} + a_{22}b_{22} + a_{32}b_{23}) + (a_{13}b_{31} + a_{23}b_{32} + a_{33}b_{33})$$

$$= a_{i1}b_{1i} + a_{i2}b_{2i} + a_{i3}b_{3i}$$

$$= a_{ij}b_{ji} \quad (n = 3)$$

(b)

$$(g_{11}^1 + g_{12}^1 + g_{21}^1 + g_{22}^1 + g_{11}^2 + g_{12}^2 + g_{21}^2 + g_{22}^2)$$

$$= g_{1i}^1 C_i + g_{2i}^1 C_i + g_{1i}^2 C_i + g_{2i}^2 C_i$$

$$= (g_{ji}^1 C_j) C_i + (g_{ji}^2 C_j) C_i$$

$$[C_i, C_j = 1 \quad \forall i, j = 1, 2]$$

$$= (g_{ji}^k C_k) C_j C_i = g_{ji}^k C_k C_j C_i$$

$$(C_k = C_j = C_i = 1) \quad (i, j, k = 1, 2) \quad \text{i.e. } \underline{(n = 2)}$$

Exercise 1.6

(1.6) For $n = 2$ write out explicitly the tripple summation

$$C_{rst} x^r y^s z^t$$

$$C_{rst} x^r y^s z^t = C_{rs1} x^r y^s z^1 + C_{rs2} x^r y^s z^2$$

$$= C_{r11} x^r y^1 z^1 + C_{r21} x^r y^2 z^1 + C_{r12} x^r y^1 z^2 + C_{r22} x^r y^2 z^2$$

$$= C_{111} x^1 y^1 z^1 + C_{211} x^2 y^1 z^1 + C_{121} x^1 y^2 z^1 + C_{221} x^2 y^2 z^1$$

$$+ C_{112} x^1 y^1 z^2 + C_{212} x^2 y^1 z^2 + C_{122} x^1 y^2 z^2 + C_{222} x^2 y^2 z^2$$

Exercise 1.7

(1.7) show that $a_{ij}x_i x_j = 0$ if $a_{ij} = i - j$

$$\begin{aligned}(i - j)x_i x_j &= (i - 1)x_i x_1 + (i - 2)x_i x_2 \\&= (2 - 1)x_2 x_1 + (1 - 1)x_1 x_1 + (1 - 2)x_1 x_2 + (2 - 2)x_2 x_2 \\&= x_2 x_1 + 0 + -x_1 x_2 + 0 \\&= 0\end{aligned}$$

Exercise 1.8

(1.8) If a_{ij} are const. calculate partial derivatives

$$\begin{aligned}\frac{\partial}{\partial x_K} (a_{ij}x_i x_j) &= a_{ij} \frac{\partial}{\partial x_K} x_i x_j \\&= a_{ij} \partial_{x_K} x_i x_j \\&= a_{ij} (\delta_{iK} x_j + \delta_{Kj} x_i) \\&= a_{ij} \delta_{iK} x_j + a_{ij} \delta_{Kj} x_i \\&= a_{Kj} x_j + a_{iK} x_i\end{aligned}$$

Since i, j has same range,

$$\frac{\partial}{\partial x_K} (a_{ij}x_i x_j) = (a_{Ki} + a_{iK}) x_i$$

Exercise 1.9

(1.9) Substitutions, kronecker delta :-

express $b^{ij}y_i y_j$ in Terms of x -variables, if $y_i = C_{ij}x_j$

& $b^{ij}C_{jk} = \delta_k^i$

$$b^{ij}y_i y_j = ?$$

$$y_i = C_{ij}x_j \quad (\text{but } j \text{ is duplicated})$$

replace $j \rightarrow r$

$$y_i = C_{ir}x_r$$

$$y_j = C_{jk}x_k$$

$$b^{ij} (C_{ir}x_r) (C_{jk}x_k) = \underbrace{b^{ij} C_{ir}} C_{jk} x_r x_k$$

$$= \delta_r^j C_{jk} x_r x_k$$

$$= C_{rk} x_r x_k$$

$$b^{ij} y_i y_j = C_{rk} x_r x_k$$

$$= C_{ij} x_i x_j$$

$[(i, j, r, k)]$ has *same range*

$$(1 \leq r, k, i, j \leq n)$$

Exercise 1.11

(1.11) If $a_{ij} = a_{ji}$

$$\begin{aligned} & \frac{\partial^2}{\partial x_K \partial x_\ell} (a_{ij} x_i x_j) \\ &= \frac{\partial}{\partial x_K} \left(\frac{\partial}{\partial x_\ell} (a_{ij} x_i x_j) \right) \\ &= \frac{\partial}{\partial x_K} (a_{ij} \delta_{i\ell} x_j + a_{ij} \delta_{j\ell} x_i) \\ &= \frac{\partial}{\partial x_K} (a_{\ell j} x_j + a_{i\ell} x_i) \\ &= \frac{\partial}{\partial x_K} ((a_{\ell i} + a_{i\ell}) x_i) \\ &= (a_{\ell i} + a_{i\ell}) \delta_{iK} \\ &= a_{\ell K} + a_{K\ell} = 2a_{K\ell} \end{aligned}$$

Exercise 1.12

(1.12) consider a system of linear eqn of form $y^i = a^{ij}x_j$ and suppose that (b_{ij}) is matrix of numbers such that $\forall i, j \ b_{ir}a^{rj} = \delta_i^j$ (b_{ij} is inverse of a^{ij})

$$y^i = a^{ij}x_j$$

$$b_{ki}y^i = b_{ki}a^{ij}x_j = \delta_k^j x_j$$

$$b_{ji}y^i = x_j$$

$$x_j = b_{ji}y^i$$

or

$$x_i = b_{ij}y^j$$

$$b_{ki}y^i = x_k$$

or

$$(x_i = b_{ij}y^j)$$

[Both represents same solⁿ.]

Exercise 1.13

(1.13) show that,

$$a_{ijk}(x_i + y_j)z_k \neq a_{ijk}x_iz_k + a_{ijk}y_jz_k$$

[i, j, k = dummy indices ,]

[$\uparrow j$ = free ; $\uparrow i$ = free]

So *not equal*.

Supplementary problems will be addressed in revision (in the end of all 13-chapters).

2 Chapter 2: Basic Linear Algebra for Tensors

[Page heading as written : (Chapter -2) (Basic Linear Algebra for Tensors)]

(2.1) Introduction :-

[A struck-out label "(1.15)" precedes "(2.1)" in the margin.]

The main purpose is to reformulate the expressions of linear algebra and matrix theory using summation convention.

(2.2) Tensor notation for matrixes, vectors & determinants

(a_{ij}) represents an element of matrix

$i \rightarrow$ row number (1 to m)

$j \rightarrow$ column number (1 to n)

A fuller notation is $[a_{ij}]_{mn}$

upper index matrix notation

$$[a_j^i]_{mn} \equiv \begin{bmatrix} a_1^1 & a_2^1 & a_3^1 & \cdots & a_n^1 \\ a_1^2 & a_2^2 & a_3^2 & \cdots & a_n^2 \\ a_1^3 & a_2^3 & a_3^3 & \cdots & a_n^3 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_1^m & a_2^m & a_3^m & \cdots & a_n^m \end{bmatrix}$$

$$[a^{ij}]_{mn} \equiv \begin{bmatrix} a^{11} & a^{12} & a^{13} & \cdots & a^{1n} \\ a^{21} & a^{22} & a^{23} & \cdots & a^{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a^{m1} & a^{m2} & a^{m3} & \cdots & a^{mn} \end{bmatrix}$$

(ex 2.1)

$$[c_j^i]_{23} = \begin{bmatrix} c_1^1 & c_2^1 & c_3^1 \\ c_1^2 & c_2^2 & c_3^2 \end{bmatrix}$$

$$[d_i^j]_{23} = \begin{bmatrix} d_1^1 & d_2^1 & d_3^1 \\ d_1^2 & d_2^2 & d_3^2 \end{bmatrix} \equiv [d_j^i]_{23}$$

$$[x_s^r]_{14} = \begin{bmatrix} x_1^1 & x_2^1 & x_3^1 & x_4^1 \end{bmatrix}$$

$$[y^{pq}]_{42} = \begin{bmatrix} y^{11} & y^{12} \\ y^{21} & y^{22} \\ y^{31} & y^{32} \\ y^{41} & y^{42} \end{bmatrix}$$

Vectors :-

A real n -dimensional vector is any column matrix $V = [x_{ij}]_{n1}$ with real components ($x_i = x_{i1}$) (or) we can simply write ($V = (x_i)$). The collection of all real n -dimensional vectors is the n -dimensional real vector space denoted R^n .

(i.e. R^n is made up of all real n-d vectors.)

Vector sums are determined by coordinate wise addⁿ as are matrix sums.

if $A \equiv [a_{ij}]_{mn}$ & $B \equiv [b_{ij}]_{mn}$

Then $A + B \equiv [a_{ij} + b_{ij}]_{mn}$

Scalar multiplication of a vector or matrix is defined by $\lambda[a_{ij}]_{mn} \equiv [\lambda a_{ij}]_{mn}$

Basic Formulas

(1) matrix multiplication

$A \equiv [a_{ij}]_{mn}$ & $B \equiv [b_{ij}]_{nk}$ then

$$AB = [a_{ir}b_{rj}]_{mk}$$

$\hookrightarrow$ represents summation over 'r'.

For mixed / upper indices :-

$$\begin{array}{l|l} AB \equiv [a_j^i]_{mn} [b_j^i]_{nk} & AB \equiv [a^{ij}]_{mn} [b^{ij}]_{nk} \\ AB = [a_r^i b_j^r]_{mk} & AB = [a^{ir} b^{sj}]_{mk} \end{array}$$

(2) Identity matrix

In terms of kronecker delta, identity matrix of order 'n' is

$$I = [\delta_{ij}]_{nn} \equiv [\delta_j^i]_{nn} = [\delta^{ij}]_{nn}$$

which has property ($IA = AI = A$) for $A = [a^{ij}]_{nn}$

or $[a_{ij}]_{nn}$

or $[a_j^i]_{nn}$

(3) Inverse of a square matrix :-

A square matrix $A = [a_{ij}]_{nn}$ is invertible if $\exists$ a unique matrix $B = [b_{ij}]_{nn}$ called inverse of A such that ($AB = BA = I$)

In terms of components criterion reads

$$\begin{aligned} AB &= BA = I \\ (a_{ir}b_{rj} &= b_{ir}a_{rj} = \delta_{ij}) \end{aligned}$$

(or) For mixed or upper indices

$$\begin{aligned} a_r^i b_j^r &= b_r^i a_j^r = \delta_j^i \\ a^{ir} b^{rj} &= b^{ir} a^{rj} = \delta^{ij} \end{aligned}$$

(4) Transpose of a matrix :-

Transpose of arbitrary matrix is

$$A^T \equiv [a_{ij}]_{mn}^T = [a_{ji}]_{nm}$$

if $a_{ij} = a_{ji}$ (A is symmetric matrix)

$$a_{ij} = -a_{ji} \quad A \text{ is skew-symmetric}$$

or Anti^o-symmetric

(5) orthogonal matrix :-

A matrix ' A ' is orthogonal if $A^T = A^{-1}$

or if $A^T A = AA^T = I$

$$\begin{aligned} (A^T = A^{-1}) \quad A &= [a_{ij}]_{nn} \\ a_{jp} &= b_{ij} \quad (A^{-1}) = b_{ij} \end{aligned}$$

Permutation symbol :

$$\begin{aligned} e_{ijk\cdots\omega} &= 0 \quad (\text{if any two subscripts are same}) \\ &= (-1)^P \end{aligned}$$

P = no. of subscript transpositions (interchg) required to bring $(i, j, k \cdots \omega)$ to natural order $(1, 2, 3, \cdots n)$

determinent of square matrix :-

If $A \equiv [a_{ij}]_{nn}$ is any square matrix

then

$$\det A = e_{i_1 i_2 i_3 \cdots i_n} a_{1i_1} a_{2i_2} a_{3i_3} \cdots a_{ni_n}$$

(ex) for 2×2 matrix

$$\begin{aligned} A &= \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} \\ |A| &= e_{i_1 i_2} a_{1i_1} a_{2i_2} \end{aligned}$$

repeated indices get summed over

So,

$$|A| = e_{1i_2} a_{11} a_{2i_2} + e_{2i_2} a_{12} a_{2i_2}$$

$$\begin{aligned}
&= a_{11} (e_{1i_2} a_{2i_2}) + a_{12} (e_{2i_2} a_{2i_2}) \\
&= a_{11} \left(\underbrace{e_{11}}_0 a_{21} + e_{12} a_{22} \right) + a_{12} \left(e_{21} a_{21} + \underbrace{e_{22}}_0 a_{22} \right) \\
&= e_{12} a_{11} a_{22} + e_{21} a_{12} a_{21} \\
|A| &= a_{11} a_{22} - a_{12} a_{21}
\end{aligned}$$

$$|AB| = |A||B| \quad \text{and} \quad |A^T| = |A|$$

laplace expansion of a determinant :-

let M_{ij} be minor of a_{ij} in A and C_{ij} be cofactor of a_{ij} in A .

$$\begin{aligned}
C_{ij} &= (-1)^{i+j} M_{ij} \\
|A| &= a_{1j} C_{1j} = a_{2j} C_{2j} = a_{3j} C_{3j} = \cdots a_{nj} C_{nj} \\
|A| &= a_{i1} C_{i1} = a_{i2} C_{i2} = a_{i3} C_{i3} \cdots = a_{in} C_{in}
\end{aligned}$$

→ repeated indices gets summed over

$$\boxed{|A| = a_{11} C_{11} + a_{12} C_{12} + a_{13} C_{13}}$$

Scalar product of vectors :-

If $u = (x_i)$

$v = (y_i)$

$$\begin{aligned}
u \cdot v &= u^T v = x_i y_i = x_1 y_1 + x_2 y_2 + x_3 y_3 + \cdots + x_n y_n \\
u^2 &= u \cdot u = x_i x_i \\
u \cdot v &= 0 \Rightarrow \bar{u} \text{ \& } \bar{v} \text{ are orthogonal. } (x_i y_i = 0)
\end{aligned}$$

norm/length of a vector :-

for $\bar{u} = (x_i)$

$$\|u\| = \sqrt{x_i x_i}$$

angle b/w two vectors :-

angle b/w two vectors $u = (x_i)$ & $V = (y_i)$ is defined by

$$\cos \theta = \frac{uv}{|u||v|} = \frac{x_i y_i}{\sqrt{x_j x_j} \sqrt{y_k y_k}}$$

for $\theta = \pi/2$ we need $(x_i y_i = u \cdot v = 0)$.

vector product/cross product in $\mathbb{R}^3$:-

$$u = (x_i)$$

$$v = (y_i)$$

$$u \times v = \hat{i} (x_2 y_3 - x_3 y_2) - \hat{j} (x_1 y_3 - x_3 y_1) + (x_1 y_2 - y_1 x_2) \hat{k}$$

$$\boxed{(\bar{u} \times \bar{v})_i = \epsilon_{ijk} x_j y_k}$$

or

$$((\bar{u} \times \bar{v})_i = \epsilon_{ijk} x_j y_k)$$

(2.3) Inverting a matrix :-

A square matrix is invertible if there exist A^{-1} such that

$$(AA^{-1} = A^{-1}A = I)$$

overview — [⌋](#) prev